

LIN7216 – Formal Linguistics

Pronouns

Adèle Hénot-Mortier

08/10/2026

Queen Mary University of London

Brainstorming

Pronouns, uses and constraints

- What are the different kinds of pronouns in English?
- What are pronouns used for?
- Are there situations (social/syntactic/contextual) in which a pronoun can't be used?

Plan for today

- Discuss different kinds of pronouns, and introduce features.
- Discuss the different uses of pronouns, focusing on the semantics of reference.
- Define syntactic conditions governing the distribution of pronouns.

Plan for today

- Discuss different kinds of pronouns, and introduce features.
- Discuss the different uses of pronouns, focusing on the semantics of reference.
- Define syntactic conditions governing the distribution of pronouns.

Plan for today

- Discuss different kinds of pronouns, and introduce features.
- Discuss the different uses of pronouns, focusing on the semantics of reference.
- Define syntactic conditions governing the distribution of pronouns.

Different kinds of pronouns

What's a pronoun?

- *Pronoun* literally means “stands for a noun”. Pronouns are closed-class, functional elements which distribute like DPs.

(1) a.	She is smart.	Personal pronoun (subject)
b.	Jo likes him .	Personal pronoun (object)
c.	Jo called her mom.	Possessive pronoun
d.	I woke up at 11. pro Made a nice brunch.	Null pronoun (dropped)
e.	Jo tried PRO to laugh.	Null pronoun (controlled)
f.	Al trusts himself .	Reflexive pronoun
g.	The cats hate each other .	Reciprocal
h.	Al bought this .	Demonstrative pronoun
i.	Who did Jo talk to?	Interrogative pronoun
j.	Jo, who Al saw, just left the party.	Relative pronoun
k.	Someone smiled.	Indefinite pronoun

Pronouns as proforms

- Pronouns are a kind of **proform**. Non-prominental proforms can replace PPs, VPs, or even entire clauses (CPs).

- (2)
- | | |
|--|-------------|
| a. Jo lives in London , and Ed lives there too. | PP anaphora |
| b. Jo arrived at noon , and Ed arrived then too. | PP anaphora |
| c. Lu baked cookies and so did Max. | VP anaphora |
| d. Lu thinks Sam will come and I think so , too. | CP anaphora |

- **Expletives** form a special class of “dummy” pronouns, which do not refer, and in English often ensure “impersonal” sentences have an overt subject.¹

- (3)
- | | |
|--|-------------------------|
| a. There is a cat on the mat. | Locative construction |
| b. It's raining. | Weather construction |
| c. It's annoying that Jo couldn't come. | Extraposed construction |

¹A requirement called the Extended Projection Principle or EPP.

Pronouns as proforms

- Pronouns are a kind of **proform**. Non-prominental proforms can replace PPs, VPs, or even entire clauses (CPs).
- (2)
- | | | |
|----|---|-------------|
| a. | Jo lives in London , and Ed lives there too. | PP anaphora |
| b. | Jo arrived at noon , and Ed arrived then too. | PP anaphora |
| c. | Lu baked cookies and so did Max. | VP anaphora |
| d. | Lu thinks Sam will come and I think so , too. | CP anaphora |
- **Expletives** form a special class of “dummy” pronouns, which do not refer, and in English often ensure “impersonal” sentences have an overt subject.¹
- (3)
- | | | |
|----|---|-------------------------|
| a. | There is a cat on the mat. | Locative construction |
| b. | It's raining. | Weather construction |
| c. | It's annoying that Jo couldn't come. | Extraposed construction |

¹A requirement called the Extended Projection Principle or **EPP**.

Zooming in on Φ -bearing pronouns

- Today, we'll mainly focus on personal, possessive, reflexive and reciprocal pronouns.
- These pronouns overtly bear person and number features, and sometimes gender features. These features are called Φ -features.

(4)	a. I [1.SG] vs. we [1.PL]	Personal pronouns (subject)
	b. Him [3.SG] vs. them [3.PL]	Personal pronouns (object)
	c. Her [3.SG.F] vs. his [3.SG.M]	Possessives
	d. Myself [1.SG] vs. Yourself [2.SG]	Reflexives
	e. Each other [3.PL]	Reciprocal

- Syntactically, Φ -features are involved in verbal **agreement**, usually between the verb (*via* the T-head) and the subject.²
- Semantically, Φ -features **constrain the interpretation** of the referent (e.g. imply they are plural, male, etc.).

²More on that next week.

Zooming in on Φ -bearing pronouns

- Today, we'll mainly focus on personal, possessive, reflexive and reciprocal pronouns.
- These pronouns overtly bear person and number features, and sometimes gender features. These features are called **Φ -features**.

(4)	a. I [1.SG] vs. we [1.PL]	Personal pronouns (subject)
	b. Him [3.SG] vs. them [3.PL]	Personal pronouns (object)
	c. Her [3.SG.F] vs. his [3.SG.M]	Possessives
	d. Myself [1.SG] vs. Yourself [2.SG]	Reflexives
	e. Each other [3.PL]	Reciprocal

- Syntactically, Φ -features are involved in verbal **agreement**, usually between the verb (*via* the T-head) and the subject.²
- Semantically, Φ -features **constrain the interpretation** of the referent (e.g. imply they are plural, male, etc.).

²More on that next week.

Zooming in on Φ -bearing pronouns

- Today, we'll mainly focus on personal, possessive, reflexive and reciprocal pronouns.
- These pronouns overtly bear person and number features, and sometimes gender features. These features are called Φ -features.

(4)	a. I [1.SG] vs. we [1.PL]	Personal pronouns (subject)
	b. Him [3.SG] vs. them [3.PL]	Personal pronouns (object)
	c. Her [3.SG.F] vs. his [3.SG.M]	Possessives
	d. Myself [1.SG] vs. Yourself [2.SG]	Reflexives
	e. Each other [3.PL]	Reciprocal

- Syntactically, Φ -features are involved in verbal **agreement**, usually between the verb (*via* the T-head) and the subject.²
- Semantically, Φ -features **constrain the interpretation** of the referent (e.g. imply they are plural, male, etc.).

²More on that next week.

Zooming in on Φ -bearing pronouns

- Today, we'll mainly focus on personal, possessive, reflexive and reciprocal pronouns.
- These pronouns overtly bear person and number features, and sometimes gender features. These features are called **Φ -features**.

(4)	a. I [1.SG] vs. we [1.PL]	Personal pronouns (subject)
	b. Him [3.SG] vs. them [3.PL]	Personal pronouns (object)
	c. Her [3.SG.F] vs. his [3.SG.M]	Possessives
	d. Myself [1.SG] vs. Yourself [2.SG]	Reflexives
	e. Each other [3.PL]	Reciprocal

- Syntactically, Φ -features are involved in verbal **agreement**, usually between the verb (*via* the T-head) and the subject.²
- Semantically, Φ -features **constrain the interpretation** of the referent (e.g. imply they are plural, male, etc.).

²More on that next week.

Zooming in on Φ -bearing pronouns

- Today, we'll mainly focus on personal, possessive, reflexive and reciprocal pronouns.
- These pronouns overtly bear person and number features, and sometimes gender features. These features are called **Φ -features**.

(4)	a. I [1.SG] vs. we [1.PL]	Personal pronouns (subject)
	b. Him [3.SG] vs. them [3.PL]	Personal pronouns (object)
	c. Her [3.SG.F] vs. his [3.SG.M]	Possessives
	d. Myself [1.SG] vs. Yourself [2.SG]	Reflexives
	e. Each other [3.PL]	Reciprocal

- Syntactically, Φ -features are involved in verbal **agreement**, usually between the verb (*via* the T-head) and the subject.²
- Semantically, Φ -features **constrain the interpretation** of the referent (e.g. imply they are plural, male, etc.).

²More on that next week.

Bonus: strong vs. weak pronouns

- In many languages,³ personal pronouns can be more or less “deficient”.
 - **Strong** pronouns can be stressed (are phonologically “whole”), topicalized, are syntactically and semantically independent, and refer to human entities.
 - **Weak** pronouns (that I conflate with the even weaker **clitics**) cannot be stressed (phonologically weak), topicalized, are syntactically and semantically dependent.

- (5)
- | | | | |
|----|--|-------------------------------|-------------------------------------|
| a. | Lui _S / *Il _W et son frère ont ri. | ‘He and his brother laughed.’ | Coordination |
| b. | Qui a ri? Moi _S !/ *Je _W ! | ‘Who laughed? Me!’ | Short answer |
| c. | Toi _S / *Tu _W , tu as ri. | ‘You, you laughed.’ | Topicalization |
| d. | Jo me _W *lui _W presente. | ‘Jo introduces me to her.’ | Person Case Constraint ⁴ |
| e. | Jo me _W présente à elle _S . | ‘Jo introduces me to her.’ | Repaired ditransitive |

³Including many from the Germanic and Romance families, but also Slovak.

⁴Ban on certain person combinations (e.g. 1+3) affecting weak indirect- and direct-object pronouns in ditransitive clusters.

Pronouns, features, and presuppositions

- Φ -features on pronouns can be taken to introduce **presuppositions**, i.e. backgrounded semantic assumptions that must be met for the sentence to be defined.
- For instance, using *She* as subject presupposes the subject identifies as female.
- The presuppositional character of inferences is often diagnosed using the *Hey, wait a minute!* test. Presuppositions pass the test while regular entailments (“at issue” stuff) don’t.

- (6) –I saw the dean this morning. **She** seemed worried.
–Hey wait a minute! I didn’t know **the dean is a woman!**
⇒ The dean being a woman is **presupposed**.
–Hey wait a minute! I didn’t know the dean seemed worried!
⇒ The dean being worried is at issue.

Pronouns, features, and presuppositions

- Φ -features on pronouns can be taken to introduce **presuppositions**, i.e. backgrounded semantic assumptions that must be met for the sentence to be defined.
- For instance, using *She* as subject presupposes the subject identifies as female.
- The presuppositional character of inferences is often diagnosed using the *Hey, wait a minute!* test. Presuppositions pass the test while regular entailments (“at issue” stuff) don’t.

- (6) –I saw the dean this morning. **She** seemed worried.
–Hey wait a minute! I didn’t know **the dean is a woman!**
⇒ The dean being a woman is **presupposed**.
–Hey wait a minute! I didn’t know the dean seemed worried!
⇒ The dean being worried is at issue.

Pronouns, features, and presuppositions

- Φ -features on pronouns can be taken to introduce **presuppositions**, i.e. backgrounded semantic assumptions that must be met for the sentence to be defined.
- For instance, using *She* as subject presupposes the subject identifies as female.
- The presuppositional character of inferences is often diagnosed using the ***Hey, wait a minute!*** test. Presuppositions pass the test while regular entailments (“at issue” stuff) don’t.

(6) –I saw the dean this morning. **She** seemed worried.

–Hey wait a minute! I didn’t know **the dean is a woman!**

⇒ The dean being a woman is **presupposed**.

–Hey wait a minute! I didn’t know the dean seemed worried!

⇒ The dean being worried is at issue.

Pronouns, features, and presuppositions

- Φ -features on pronouns can be taken to introduce **presuppositions**, i.e. backgrounded semantic assumptions that must be met for the sentence to be defined.
- For instance, using *She* as subject presupposes the subject identifies as female.
- The presuppositional character of inferences is often diagnosed using the ***Hey, wait a minute!*** test. Presuppositions pass the test while regular entailments (“at issue” stuff) don’t.

- (6) –I saw the dean this morning. **She** seemed worried.
–Hey wait a minute! I didn’t know **the dean is a woman!**
⇒ The dean being a woman is **presupposed**.
–Hey wait a minute! I didn’t know the dean seemed worried!
⇒ The dean being worried is at issue.

Pronouns, features, and presuppositions

- Presuppositions also survive in negated/modalized sentences, questions, conditional antecedents–environments which normally tend to weaken/reverse entailments.

- (7) a. It's **not** true she is worried.
 ↪ Subject is ~~not~~ female and **not** worried.
- b. **Perhaps** she is worried.
 ↪ Subject is ~~perhaps~~ female and **perhaps** worried.
- c. Is she is worried?
 ↪ Subject is female and **we wonder if** she is worried.⁵
- d. **If** she is worried, we should worry too.
 ↪ Main subject is female and **if** she is worried we should worry too.⁶

⁵And not: we wonder if the subject is female and worried.

⁶And not: if the main subject is female and worried we should worry too.

Bonus: possessives and presuppositions

- Different presuppositions can be triggered by a variety of lexical items (*only, even, too, again, stop, clefts...*).
- Possessives also trigger specific presuppositions, on top of the ones tied to Φ -features.

- (8)
- a. Tomorrow, I must bring my alligator to the vet.
Hey wait a minute! I didn't know you had a pet alligator!
 - b. I won't need to bring my alligator to the vet.
↪ I have a pet alligator.
 - c. Perhaps I will need to bring my alligator to the vet.
↪ I have a pet alligator.
 - d. Will I need to bring my alligator to the vet?
↪ I have a pet alligator.
 - e. If I need to bring my alligator to the vet?
↪ I have a pet alligator.

- Saying X's Y presupposes X owns a Y!

Bonus: possessives and presuppositions

- Different presuppositions can be triggered by a variety of lexical items (*only, even, too, again, stop, clefts...*).
- Possessives also trigger specific presuppositions, on top of the ones tied to Φ -features.

- (8)
- a. Tomorrow, I must bring **my alligator** to the vet.
Hey wait a minute! I didn't know you had a pet alligator!
 - b. I **won't** need to bring my alligator to the vet.
~> I have a pet alligator.
 - c. **Perhaps** I will need to bring my alligator to the vet.
~> I have a pet alligator.
 - d. Will I need to bring my alligator to the vet?
~> I have a pet alligator.
 - e. **If** I need to bring my alligator to the vet?
~> I have a pet alligator.

- Saying *X's Y* presupposes *X owns a Y*!

Bonus: possessives and presuppositions

- Different presuppositions can be triggered by a variety of lexical items (*only, even, too, again, stop, clefts...*).
- Possessives also trigger specific presuppositions, on top of the ones tied to Φ -features.

- (8)
- a. Tomorrow, I must bring **my alligator** to the vet.
Hey wait a minute! I didn't know you had a pet alligator!
 - b. I **won't** need to bring my alligator to the vet.
~> I have a pet alligator.
 - c. **Perhaps** I will need to bring my alligator to the vet.
~> I have a pet alligator.
 - d. Will I need to bring my alligator to the vet?
~> I have a pet alligator.
 - e. **If** I need to bring my alligator to the vet?
~> I have a pet alligator.

- Saying *X's Y* presupposes *X owns a Y*!

Different uses of pronouns

Referential use

- The most canonical use of pronouns is to refer to an antecedent provided by the discourse (**anaphoric** use), the extra-linguistic context (**deictic** use) or the utterance's index, i.e. who speaks to who, when, and where (**indexical** use).

- (9) a. A linguist entered the room. **She**_(=the linguist) seemed puzzled. Anaphoric
- b. Jo likes **herself**_(=Jo) . Anaphoric
- c. [Looking at a puppy]. **It**_(=the puppy) is so cute! Deictic
- d. **I**_(=speaker) need **you**_(=addressee) to be on time tomorrow_(=utterance day+1). Indexical

- Let's focus on the semantics of anaphora.

Semantic modeling of reference

- We assume DPs (including pronouns and proper names) bear **indices**, like 1, 3, 8 etc. For instance, *Jo*₃, *they*₅, *herself*₈ etc. These indices are accessible to the interpretation function $\llbracket \cdot \rrbracket$.

Semantic modeling of reference

- We assume DPs (including pronouns and proper names) bear **indices**, like 1, 3, 8 etc. For instance, *Jo*₃, *they*₅, *herself*₈ etc. These indices are accessible to the interpretation function $\llbracket \cdot \rrbracket$.
- We also assume $\llbracket \cdot \rrbracket$ is parametrized by an **assignment function** g , which we write $\llbracket \cdot \rrbracket^g$.

Semantic modeling of reference

- We assume DPs (including pronouns and proper names) bear **indices**, like 1, 3, 8 etc. For instance, Jo_3 , $they_5$, $herself_8$ etc. These indices are accessible to the interpretation function $\llbracket \cdot \rrbracket$.
- We also assume $\llbracket \cdot \rrbracket$ is parametrized by an **assignment function** g , which we write $\llbracket \cdot \rrbracket^g$.
- g keeps track of a mapping between indices (1, 3, 8 etc.) and individuals (elements of the metalanguage). For instance,

$$g = \begin{bmatrix} 1 \mapsto Jo \\ 3 \mapsto Ed \\ 8 \mapsto Al \end{bmatrix}.$$

Semantic modeling of reference

- We assume DPs (including pronouns and proper names) bear **indices**, like 1, 3, 8 etc. For instance, *Jo*₃, *they*₅, *herself*₈ etc. These indices are accessible to the interpretation function $\llbracket \cdot \rrbracket$.
- We also assume $\llbracket \cdot \rrbracket$ is parametrized by an **assignment function** g , which we write $\llbracket \cdot \rrbracket^g$.
- g keeps track of a mapping between indices (1, 3, 8 etc.) and individuals (elements of the metalanguage). For instance,
$$g = \begin{bmatrix} 1 \mapsto \text{Jo} \\ 3 \mapsto \text{Ed} \\ 8 \mapsto \text{Al} \end{bmatrix}.$$
- Given the g above, what's $g(3)$? $g(4)$? $g(8)$? What's the type of $g(8)$?

Semantic modeling of reference

- We assume DPs (including pronouns and proper names) bear **indices**, like 1, 3, 8 etc. For instance, *Jo*₃, *they*₅, *herself*₈ etc. These indices are accessible to the interpretation function $\llbracket \cdot \rrbracket$.
- We also assume $\llbracket \cdot \rrbracket$ is parametrized by an **assignment function** g , which we write $\llbracket \cdot \rrbracket^g$.
- g keeps track of a mapping between indices (1, 3, 8 etc.) and individuals (elements of the metalanguage). For instance,
$$g = \begin{bmatrix} 1 \mapsto \text{Jo} \\ 3 \mapsto \text{Ed} \\ 8 \mapsto \text{Al} \end{bmatrix}.$$
- Given the g above, what's $g(3)$? $g(4)$? $g(8)$? What's the type of $g(8)$? Solutions: *Ed*, undefined, *Al* (an individual).

Interpreting pronouns

- We define a special **interpretation rule for pronouns**, sensitive to their index and the current g : let x_i be a pronoun and g an assignment function. If $g(i)$ is defined, $\llbracket x_i \rrbracket^g = g(i)$, otherwise, $\llbracket x_i \rrbracket^g$ is undefined.

Interpreting pronouns

- We define a special **interpretation rule for pronouns**, sensitive to their index and the current g : let x_i be a pronoun and g an assignment function. If $g(i)$ is defined, $\llbracket x_i \rrbracket^g = g(i)$, otherwise, $\llbracket x_i \rrbracket^g$ is undefined.

- Take $g = \begin{bmatrix} 1 \mapsto \text{Jo} \\ 3 \mapsto \text{Ed} \\ 8 \mapsto \text{Al} \end{bmatrix}$. What's $\llbracket \text{him}_3 \rrbracket^g$? $\llbracket \text{himself}_1 \rrbracket^g$?

Interpreting pronouns

- We define a special **interpretation rule for pronouns**, sensitive to their index and the current g : let x_i be a pronoun and g an assignment function. If $g(i)$ is defined, $\llbracket x_i \rrbracket^g = g(i)$, otherwise, $\llbracket x_i \rrbracket^g$ is undefined.

- Take $g = \begin{bmatrix} 1 \mapsto \text{Jo} \\ 3 \mapsto \text{Ed} \\ 8 \mapsto \text{Al} \end{bmatrix}$. What's $\llbracket \text{him}_3 \rrbracket^g$? $\llbracket \text{himself}_1 \rrbracket^g$?

$$\llbracket \text{him}_3 \rrbracket^g = g(3) = \text{Ed}.$$

Interpreting pronouns

- We define a special **interpretation rule for pronouns**, sensitive to their index and the current g : let x_i be a pronoun and g an assignment function. If $g(i)$ is defined, $\llbracket x_i \rrbracket^g = g(i)$, otherwise, $\llbracket x_i \rrbracket^g$ is undefined.

- Take $g = \begin{bmatrix} 1 \mapsto \text{Jo} \\ 3 \mapsto \text{Ed} \\ 8 \mapsto \text{Al} \end{bmatrix}$. What's $\llbracket \text{him}_3 \rrbracket^g$? $\llbracket \text{himself}_1 \rrbracket^g$?

$$\llbracket \text{him}_3 \rrbracket^g = g(3) = \text{Ed}. \quad \llbracket \text{himself}_1 \rrbracket^g = g(1) = \text{Jo}.$$

Interpreting pronouns

- We define a special **interpretation rule for pronouns**, sensitive to their index and the current g : let x_i be a pronoun and g an assignment function. If $g(i)$ is defined, $\llbracket x_i \rrbracket^g = g(i)$, otherwise, $\llbracket x_i \rrbracket^g$ is undefined.

- Take $g = \begin{bmatrix} 1 \mapsto \text{Jo} \\ 3 \mapsto \text{Ed} \\ 8 \mapsto \text{Al} \end{bmatrix}$. What's $\llbracket \text{him}_3 \rrbracket^g$? $\llbracket \text{himself}_1 \rrbracket^g$?

$\llbracket \text{him}_3 \rrbracket^g = g(3) = \text{Ed}$. $\llbracket \text{himself}_1 \rrbracket^g = g(1) = \text{Jo}$.

- We assume assignment functions can be **updated**. For instance, $g + [3 \mapsto \text{Jo}]$ updates g with the mapping $3 \mapsto \text{Jo}$.

So $g + [3 \mapsto \text{Jo}] = \begin{bmatrix} 1 \mapsto \text{Jo} \\ 3 \mapsto \text{Jo} \\ 8 \mapsto \text{Al} \end{bmatrix}$. Likewise, $g + [2 \mapsto \text{Lu}] = \begin{bmatrix} 1 \mapsto \text{Jo} \\ 2 \mapsto \text{Lu} \\ 3 \mapsto \text{Ed} \\ 8 \mapsto \text{Al} \end{bmatrix}$

Updating the assignment function

- In the case of anaphora, g gets **locally updated** by the action of index-bearing referring expressions, like proper names, and a specific rule called **Predicate Abstraction**. Consider:

(10) Jo_1 likes herself₁.

- We want to derive $\llbracket Jo_1 \text{ likes herself}_1 \rrbracket^g = 1$ iff Jo likes Jo . We have $\llbracket Jo_1 \rrbracket^g = Jo$. How to make sure herself₁ ends up pointing to Jo too? Idea:
 - Make Jo_1 introduce a pronoun x_1 coreferring with herself₁ (same index!), and acting as the liker (second argument of *like*).
 - Transform x_1 likes herself₁ into the predicate of liking oneself ($\lambda x. 1$ iff x likes x). This is Predicate Abstraction.
 - Apply this newly formed predicate to Jo_1 via Functional Application.

Updating the assignment function

- In the case of anaphora, g gets **locally updated** by the action of index-bearing referring expressions, like proper names, and a specific rule called **Predicate Abstraction**. Consider:

(10) Jo_1 likes herself₁.

- We want to derive $\llbracket Jo_1 \text{ likes herself}_1 \rrbracket^g = 1$ iff Jo likes Jo . We have $\llbracket Jo_1 \rrbracket^g = Jo$. How to make sure herself₁ ends up pointing to Jo too? Idea:
 - Make Jo_1 introduce a pronoun x_1 coreferring with herself₁ (same index!), and acting as the liker (second argument of *like*).
 - Transform x_1 likes herself₁ into the predicate of liking oneself ($\lambda x. 1$ iff x likes x). This is Predicate Abstraction.
 - Apply this newly formed predicate to Jo_1 via Functional Application.

Updating the assignment function

- In the case of anaphora, g gets **locally updated** by the action of index-bearing referring expressions, like proper names, and a specific rule called **Predicate Abstraction**. Consider:

(10) Jo_1 likes herself₁.

- We want to derive $\llbracket Jo_1 \text{ likes herself}_1 \rrbracket^g = 1$ iff Jo likes Jo . We have $\llbracket Jo_1 \rrbracket^g = Jo$. How to make sure herself₁ ends up pointing to Jo too? Idea:
 - Make Jo_1 introduce a pronoun x_1 coreferring with herself₁ (same index!), and acting as the liker (second argument of *like*).
 - Transform x_1 likes herself₁ into the predicate of liking oneself ($\lambda x. 1$ iff x likes x). This is Predicate Abstraction.
 - Apply this newly formed predicate to Jo_1 via Functional Application.

Updating the assignment function

- In the case of anaphora, g gets **locally updated** by the action of index-bearing referring expressions, like proper names, and a specific rule called **Predicate Abstraction**. Consider:

(10) Jo_1 likes herself₁.

- We want to derive $\llbracket Jo_1 \text{ likes herself}_1 \rrbracket^g = 1$ iff Jo likes Jo. We have $\llbracket Jo_1 \rrbracket^g = Jo$. How to make sure herself₁ ends up pointing to Jo too? Idea:
 - Make Jo_1 introduce a pronoun x_1 coreferring with herself₁ (same index!), and acting as the liker (second argument of *like*).
 - Transform x_1 likes herself₁ into the predicate of liking oneself ($\lambda x. 1$ iff x likes x). This is Predicate Abstraction.
 - Apply this newly formed predicate to Jo_1 via Functional Application.

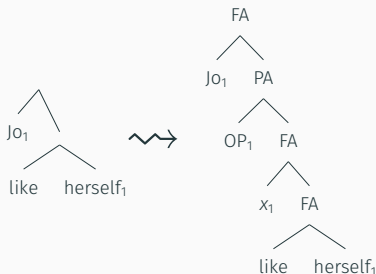
Updating the assignment function

- In the case of anaphora, g gets **locally updated** by the action of index-bearing referring expressions, like proper names, and a specific rule called **Predicate Abstraction**. Consider:

(10) Jo_1 likes herself₁.

- We want to derive $\llbracket Jo_1 \text{ likes herself}_1 \rrbracket^g = 1$ iff Jo likes Jo . We have $\llbracket Jo_1 \rrbracket^g = Jo$. How to make sure herself₁ ends up pointing to Jo too? Idea:
 - Make Jo_1 introduce a pronoun x_1 coreferring with herself₁ (same index!), and acting as the liker (second argument of *like*).
 - Transform x_1 likes herself₁ into the predicate of liking oneself ($\lambda x. 1$ iff x likes x). This is Predicate Abstraction.
 - Apply this newly formed predicate to Jo_1 via Functional Application.

Coreference via Predicate Abstraction



- **Predicate Abstraction:**

$$\llbracket OP_i \beta \rrbracket^g = \lambda x. \llbracket \beta \rrbracket^{g+[i \mapsto x]}.$$
- Creates a predicate $\lambda x...$ by replacing all the i -indexed pronouns in β with x .

- For any g , $\llbracket x_1 \text{ like herself}_1 \rrbracket^g = 1$ iff $g(1)$ likes $g(1)$
- Now applying predicate abstraction:

$$\llbracket OP_1 x_1 \text{ like herself}_1 \rrbracket^{g'} = \lambda x. \llbracket x_1 \text{ like herself}_1 \rrbracket^{g'+[1 \mapsto x]}$$

- Let's define $g = g' + [1 \mapsto x]$. We have $g(1) = (g' + [1 \mapsto x])(1) = x$.

$$\llbracket OP_1 x_1 \text{ like herself}_1 \rrbracket^{g'} = \lambda x. 1 \text{ iff } g(1) \text{ likes } g(1) = \lambda x. 1 \text{ iff } x \text{ likes } x$$

$$\llbracket OP_1 x_1 \text{ like herself}_1 \rrbracket^{g'} (\llbracket Jo_1 \rrbracket^{g'}) = (\lambda x. 1 \text{ iff } x \text{ likes } x)(Jo)$$

$$\llbracket OP_1 x_1 \text{ like herself}_1 \rrbracket^{g'} (\llbracket Jo_1 \rrbracket^{g'}) = 1 \text{ iff } Jo \text{ likes } Jo$$

Bonus: strict/sloppy ambiguity

- Let's combine pronominal anaphora and **ellipsis**:

(11) Jo₁ likes her₁ mom and Al₂ does ~~like x₁'s mom~~ too.

- The first part of the sentence has a clear meaning: Jo likes Jo's mom. What about the second part?
 - It can mean Al likes Al's mom ("**sloppy**" reading);
 - It can also mean Al likes Jo's mom ("**strict**" reading).
- Very roughly, the sloppy reading corresponds to a pronoun that is interpreted relative to the variable introduced by abstraction in the elided clause, whereas the strict reading corresponds to a pronoun whose reference is fixed before abstraction.

Bonus: strict/sloppy ambiguity

- Let's combine pronominal anaphora and **ellipsis**:

(11) Jo₁ likes her₁ mom and Al₂ does ~~like x_i's mom~~ too.

- The first part of the sentence has a clear meaning: Jo likes Jo's mom. What about the second part?
 - It can mean Al likes Al's mom ("**sloppy**" reading);
 - It can also mean Al likes Jo's mom ("**strict**" reading).
- Very roughly, the sloppy reading corresponds to a pronoun that is interpreted relative to the variable introduced by abstraction in the elided clause, whereas the strict reading corresponds to a pronoun whose reference is fixed before abstraction.

Bonus: strict/sloppy ambiguity

- Let's combine pronominal anaphora and **ellipsis**:

(11) Jo₁ likes her₁ mom and Al₂ does ~~like x_i's mom~~ too.

- The first part of the sentence has a clear meaning: Jo likes Jo's mom. What about the second part?
 - It can mean Al likes Al's mom ("sloppy" reading);
 - It can also mean Al likes Jo's mom ("strict" reading).
- Very roughly, the sloppy reading corresponds to a pronoun that is interpreted relative to the variable introduced by abstraction in the elided clause, whereas the strict reading corresponds to a pronoun whose reference is fixed before abstraction.

Bonus: strict/sloppy ambiguity

- Let's combine pronominal anaphora and **ellipsis**:

(11) Jo₁ likes her₁ mom and Al₂ does ~~like x_i's mom~~ too.

- The first part of the sentence has a clear meaning: Jo likes Jo's mom. What about the second part?
 - It can mean Al likes Al's mom ("**sloppy**" reading);
 - It can also mean Al likes Jo's mom ("**strict**" reading).
- Very roughly, the sloppy reading corresponds to a pronoun that is interpreted relative to the variable introduced by abstraction in the elided clause, whereas the strict reading corresponds to a pronoun whose reference is fixed before abstraction.

Bonus: strict/sloppy ambiguity

- Let's combine pronominal anaphora and **ellipsis**:

(11) Jo₁ likes her₁ mom and Al₂ does ~~like x_i's mom~~ too.

- The first part of the sentence has a clear meaning: Jo likes Jo's mom. What about the second part?
 - It can mean Al likes Al's mom ("**sloppy**" reading);
 - It can also mean Al likes Jo's mom ("**strict**" reading).
- Very roughly, the sloppy reading corresponds to a pronoun that is interpreted relative to the variable introduced by abstraction in the elided clause, whereas the strict reading corresponds to a pronoun whose reference is fixed before abstraction.

Bonus: strict/sloppy ambiguity

- Let's combine pronominal anaphora and **ellipsis**:

(11) Jo₁ likes her₁ mom and Al₂ does ~~like x_i's mom~~ too.

- The first part of the sentence has a clear meaning: Jo likes Jo's mom. What about the second part?
 - It can mean Al likes Al's mom ("**sloppy**" reading);
 - It can also mean Al likes Jo's mom ("**strict**" reading).
- Very roughly, the sloppy reading corresponds to a pronoun that is interpreted relative to the variable introduced by abstraction in the elided clause, whereas the strict reading corresponds to a pronoun whose reference is fixed before abstraction.

Bonus: Indexicals

- Indexical pronouns are picked up from the utterance's context (index). They typically refer to the speaker, the addressee, or related the here and now.

(12) I want you to be on time tomorrow.

The speaker wants the addressee to be on time the day after that of the utterance.

- If the context of the utterance changes, the referents of the indexicals change too! This is done by adding a new parameter to $\llbracket . \rrbracket$, c , for utterance context.
- We then have $\llbracket I \rrbracket^c$ = the speaker in c , $\llbracket you \rrbracket^c$ = the addressee in c , $\llbracket now \rrbracket^c$ = the time in c , $\llbracket here \rrbracket^c$ = the location in c etc.

Bonus: Indexicals

- Indexical pronouns are picked up from the utterance's context (index). They typically refer to the speaker, the addressee, or related the here and now.

(12) I want you to be on time tomorrow.

The speaker wants the addressee to be on time the day after that of the utterance.

- If the context of the utterance changes, the referents of the indexicals change too! This is done by adding a new parameter to $\llbracket \cdot \rrbracket$, c , for utterance context.
- We then have $\llbracket I \rrbracket^c = \text{the speaker in } c$, $\llbracket \text{you} \rrbracket^c = \text{the addressee in } c$, $\llbracket \text{now} \rrbracket^c = \text{the time in } c$, $\llbracket \text{here} \rrbracket^c = \text{the location in } c$ etc.

Bonus: Indexicals

- Indexical pronouns are picked up from the utterance's context (index). They typically refer to the speaker, the addressee, or related the here and now.

(12) I want you to be on time tomorrow.

The speaker wants the addressee to be on time the day after that of the utterance.

- If the context of the utterance changes, the referents of the indexicals change too! This is done by adding a new parameter to $\llbracket . \rrbracket$, c , for utterance context.
- We then have $\llbracket I \rrbracket^c = \text{the speaker in } c$, $\llbracket \text{you} \rrbracket^c = \text{the addressee in } c$, $\llbracket \text{now} \rrbracket^c = \text{the time in } c$, $\llbracket \text{here} \rrbracket^c = \text{the location in } c$ etc.

Bonus: Indexicals

- Indexical pronouns are picked up from the utterance's context (index). They typically refer to the speaker, the addressee, or related the here and now.

(12) I want you to be on time tomorrow.

The speaker wants the addressee to be on time the day after that of the utterance.

- If the context of the utterance changes, the referents of the indexicals change too! This is done by adding a new parameter to $\llbracket . \rrbracket$, c , for utterance context.
- We then have $\llbracket I \rrbracket^c$ = the speaker in c , $\llbracket you \rrbracket^c$ = the addressee in c , $\llbracket now \rrbracket^c$ = the time in c , $\llbracket here \rrbracket^c$ = the location in c etc.

Bonus: indexical shift languages

- In English, saying *Jo says that I am happy* means that Jo says that the speaker of the utterance *Jo says that...* is happy. This suggests the context parameter *c* remains set to the global utterance context, even under *verba dicendi* like *say*.
- But in some languages,⁷ embedded indexicals can pick up referents based on the local, embedded utterance context. So, *Jo says that I am happy* can mean *Jo says that Jo is happy*.
- This is called **indexical shift**, shown below in Uyghur.

- (13) a. Ahmet [men ket-tim] di-di.
Ahmet [1.SG leave-PST.1.SG] say-PST.3
'Ahmeti said that he, the speaker of embedded sentence, left.'
(shifted reading)
- b. Ahmet [meni ket-ti] di-di.
Ahmet [1.SG.ACC leave-PST.3] say-PST.3
'Ahmet said that I, the speaker of the global utterance, left.'
(non-shifted reading)

⁷Amharic, Salve, Zazaki, Uyghur...

Bonus: indexical shift languages

- In English, saying *Jo says that I am happy* means that Jo says that the speaker of the utterance *Jo says that...* is happy. This suggests the context parameter *c* remains set to the global utterance context, even under *verba dicendi* like *say*.
- But in some languages,⁷ embedded indexicals can pick up referents based on the local, embedded utterance context. So, *Jo says that I am happy* can mean *Jo says that Jo is happy*.
- This is called **indexical shift**, shown below in Uyghur.

- (13) a. Ahmet [men ket-tim] di-di.
Ahmet [1.SG leave-PST.1.SG] say-PST.3
'Ahmeti said that he, the speaker of embedded sentence, left.'
(shifted reading)
- b. Ahmet [meni ket-ti] di-di.
Ahmet [1.SG.ACC leave-PST.3] say-PST.3
'Ahmet said that I, the speaker of the global utterance, left.'
(non-shifted reading)

⁷Amharic, Salve, Zazaki, Uyghur...

- Many language use specific pronouns, called **honorifics**, to refer to individuals politely.
- In many cases still, honorifics are not dedicated forms, but instead recycle existing non-polite Φ -features. Very often, plural/third person are recruited, and reflected by agreement on the verb.

(14) a. French:

Vous êtes fort sympathiques.

2.PL be.2.PL very nice.PL.

(Politely) 'You are very nice.'

b. Italian:

Lei è molto gentile.

3.SG.F be.3.SG very kind.

(Politely) 'You are very kind.'

c. German:

Sie sind sehr nett.

3.PL be.3.PL very nice.

Polite use

- Many languages use specific pronouns, called **honorifics**, to refer to individuals politely.
- In many cases still, honorifics are not dedicated forms, but instead recycle existing non-polite Φ -features. Very often, plural/third person are recruited, and reflected by agreement on the verb.

(14) a. French:

Vous êtes fort sympathiques.

2.PL be.2.PL very nice.PL.

(Politely) 'You are very nice.'

b. Italian:

Lei è molto gentile.

3.SG.F be.3.SG very kind.

(Politely) 'You are very kind.'

c. German:

Sie sind sehr nett.

3.PL be.3.PL very nice.

Polite use

- Many language use specific pronouns, called **honorifics**, to refer to individuals politely.
- In many cases still, honorifics are not dedicated forms, but instead recycle existing non-polite Φ -features. Very often, plural/third person are recruited, and reflected by agreement on the verb.

(14) a. French:

Vous êtes fort sympathiques.

2.PL be.2.PL very nice.PL.

(Politely) 'You are very nice.'

b. Italian:

Lei è molto gentile.

3.SG.F be.3.SG very kind.

(Politely) 'You are very kind.'

c. German:

Sie sind sehr nett.

3.PL be.3.PL very nice.

Bound variable use

- Sometimes, pronouns do not really refer to a (fixed) referent. This happens when the pronoun is **quantified** over, via expressions like *No student*, *Each student*, *Every farmer*, called quantifier phrases (QPs).

- (15) **No** student believes **they** will pass. Bound (personal) pronoun
- (16) **Each** student called **their** parents. Bound (possessive) pronoun
- (17) **Every** farmer who owns a donkey treats **it** well. Bound(??⁸) (personal) pronoun

- Deriving the meaning of these quantified sentences builds on Predicate Abstraction and will be the object of an optional lecture later this semester.

⁸Structures like (17), called *donkey-sentences*, are puzzles for simple theories of anaphora, as the QP *a donkey* does not C-Command the bound pronoun *it*, i.e. does not syntactically binds *it*, yet, co-reference seems to be established.

Bound variable use

- Sometimes, pronouns do not really refer to a (fixed) referent. This happens when the pronoun is **quantified** over, via expressions like *No student*, *Each student*, *Every farmer*, called quantifier phrases (QPs).

- (15) **No** student believes **they** will pass. Bound (personal) pronoun
- (16) **Each** student called **their** parents. Bound (possessive) pronoun
- (17) **Every** farmer who owns a donkey treats **it** well. Bound(??⁸) (personal) pronoun

- Deriving the meaning of these quantified sentences builds on Predicate Abstraction and will be the object of an optional lecture later this semester.

⁸Structures like (17), called *donkey-sentences*, are puzzles for simple theories of anaphora, as the QP *a donkey* does not C-Command the bound pronoun *it*, i.e. does not syntactically binds *it*, yet, co-reference seems to be established.

Bound variable use

- Sometimes, pronouns do not really refer to a (fixed) referent. This happens when the pronoun is **quantified** over, *via* expressions like *No student*, *Each student*, *Every farmer*, called quantifier phrases (QPs).

(15) **No** student believes **they** will pass. Bound (personal) pronoun

(16) **Each** student called **their** parents. Bound (possessive) pronoun

(17) **Every** farmer who owns a donkey treats **it** well. Bound(??⁸) (personal) pronoun

- Deriving the meaning of these quantified sentences builds on Predicate Abstraction and will be the object of an optional lecture later this semester.

⁸Structures like (17), called *donkey-sentences*, are puzzles for simple theories of anaphora, as the QP *a donkey* does not C-Command the bound pronoun *it*, i.e. does not syntactically binds *it*, yet, co-reference seems to be established.

Null pronouns

- In many languages, incl. Italian, Turkish, Japanese, Standard Arabic, Hungarian, Swahili, subjects need not be overt.

(18) Sen-i gör-dü-m
 2.SG.ACC see.PST.1.SG
 ‘(I) saw you.’

- In many cases, the pronoun can be **recovered** based on the **agreement marking** on the verb (e.g. above, 1.SG recovers *I*). If not, contextual cues can help too.
- Languages allowing null subjects are called *pro-drop*; but in many frameworks including Minimalism, they are not really dropping anything in the syntax, for they are assumed to involve a covert pronoun (*pro*) in Spec-TP.

- In many languages, incl. Italian, Turkish, Japanese, Standard Arabic, Hungarian, Swahili, subjects need not be overt.

(18) Sen-i gör-dü-m
 2.SG.ACC see.PST.1.SG
 ‘(I) saw you.’

- In many cases, the pronoun can be **recovered** based on the **agreement marking** on the verb (e.g. above, 1.SG recovers *I*). If not, contextual cues can help too.
- Languages allowing null subjects are called ***pro-drop***; but in many frameworks including Minimalism, they are not really dropping anything in the syntax, for they are assumed to involve a covert pronoun (*pro*) in Spec-TP.

Control constructions

- **Control** constructions involving verbs like *want*, *plan*, *try*, followed by an **infinitival clause**, are assumed to involve a null embedded subject, glossed PRO (“big PRO”), which is “controlled” by (co-referential with) the main subject.⁹
- Intuitively, control predicates assign an AGENT role to their subject, and embed predicates assigning a potentially different role to PRO.
- This makes them incompatible with expletive subjects and split idioms.¹⁰

- (19) a. Jo wants PRO to leave
b. * It/there wants that Jo leaves.
c. * The cat tried PRO to be out of the bag.

⁹In so-called *partial control* constructions, PRO refers to a plural individual strictly containing the subject: *The chair wants PRO to gather at 6.*

¹⁰Compare with so-called raising predicates, like *seem*!

Control constructions

- **Control** constructions involving verbs like *want*, *plan*, *try*, followed by an **infinitival clause**, are assumed to involve a null embedded subject, glossed PRO (“big PRO”), which is “controlled” by (co-referential with) the main subject.⁹
- Intuitively, control predicates assign an AGENT role to their subject, and embed predicates assigning a potentially different role to PRO.
- This makes them incompatible with expletive subjects and split idioms.¹⁰

- (19) a. Jo wants PRO to leave
b. * It/there wants that Jo leaves.
c. * The cat tried PRO to be out of the bag.

⁹In so-called *partial control* constructions, PRO refers to a plural individual strictly containing the subject: *The chair wants PRO to gather at 6.*

¹⁰Compare with so-called raising predicates, like *seem*!

Control constructions

- **Control** constructions involving verbs like *want*, *plan*, *try*, followed by an **infinitival clause**, are assumed to involve a null embedded subject, glossed PRO (“big PRO”), which is “controlled” by (co-referential with) the main subject.⁹
- Intuitively, control predicates assign an AGENT role to their subject, and embed predicates assigning a potentially different role to PRO.
- This makes them incompatible with expletive subjects and split idioms.¹⁰

- (19) a. Jo wants PRO to leave
b. * It/there wants that Jo leaves.
c. * The cat tried PRO to be out of the bag.

⁹In so-called *partial control* constructions, PRO refers to a plural individual strictly containing the subject: *The chair wants PRO to gather at 6.*

¹⁰Compare with so-called raising predicates, like *seem*!

Constraints on pronouns

When to use which pronoun

- Let us further investigate the syntactic conditions governing the distribution of:
 - personal pronouns (now just “**pronouns**”);
 - reflexives/reciprocals (called “**anaphors**”);
 - as well as referring expressions (“**R-expressions**”)—typically proper names and full DPs.
- These elements will be indexed with letters like *i* and *j*. If two elements from the same sentence are both indexed with say, *i*, it means they **corefer** (have same referent). If they are indexed with, say, *i* and *j*, we assume they have different referents.

(20) Jo_{*i*} said that they_{*j*}... Coreferring R-expression and pronoun

(21) He_{*i*} told himself_{*j*}... Coreferring pronoun and anaphor

(22) She_{*i*} told her_{*j*}... Non-coreferring pronouns

- Under which syntactic configuration(s) can pronouns, anaphors, and R-expressions corefer?

When to use which pronoun

- Let us further investigate the syntactic conditions governing the distribution of:
 - personal pronouns (now just “**pronouns**”);
 - reflexives/reciprocals (called “**anaphors**”);
 - as well as referring expressions (“**R-expressions**”)—typically proper names and full DPs.
- These elements will be indexed with letters like *i* and *j*. If two elements from the same sentence are both indexed with say, *i*, it means they **corefer** (have same referent). If they are indexed with, say, *i* and *j*, we assume they have different referents.

(20) Jo_{*i*} said that they_{*j*}... Coreferring R-expression and pronoun

(21) He_{*i*} told himself_{*j*}... Coreferring pronoun and anaphor

(22) She_{*i*} told her_{*j*}... Non-coreferring pronouns

- Under which syntactic configuration(s) can pronouns, anaphors, and R-expressions corefer?

When to use which pronoun

- Let us further investigate the syntactic conditions governing the distribution of:
 - personal pronouns (now just “**pronouns**”);
 - reflexives/reciprocals (called “**anaphors**”);
 - as well as referring expressions (“**R-expressions**”)—typically proper names and full DPs.
- These elements will be indexed with letters like *i* and *j*. If two elements from the same sentence are both indexed with say, *i*, it means they **corefer** (have same referent). If they are indexed with, say, *i* and *j*, we assume they have different referents.

(20) Jo_{*i*} said that they_{*j*}... Coreferring R-expression and pronoun

(21) He_{*i*} told himself_{*j*}... Coreferring pronoun and anaphor

(22) She_{*i*} told her_{*j*}... Non-coreferring pronouns

- Under which syntactic configuration(s) can pronouns, anaphors, and R-expressions corefer?

When to use which pronoun

- Let us further investigate the syntactic conditions governing the distribution of:
 - personal pronouns (now just “**pronouns**”);
 - reflexives/reciprocals (called “**anaphors**”);
 - as well as referring expressions (“**R-expressions**”)—typically proper names and full DPs.
- These elements will be indexed with letters like *i* and *j*. If two elements from the same sentence are both indexed with say, *i*, it means they **corefer** (have same referent). If they are indexed with, say, *i* and *j*, we assume they have different referents.

(20) Jo_{*i*} said that they_{*j*}... Coreferring R-expression and pronoun

(21) He_{*i*} told himself_{*j*}... Coreferring pronoun and anaphor

(22) She_{*i*} told her_{*j*}... Non-coreferring pronouns

- Under which syntactic configuration(s) can pronouns, anaphors, and R-expressions corefer?

When to use which pronoun

- Let us further investigate the syntactic conditions governing the distribution of:
 - personal pronouns (now just “**pronouns**”);
 - reflexives/reciprocals (called “**anaphors**”);
 - as well as referring expressions (“**R-expressions**”)—typically proper names and full DPs.
- These elements will be indexed with letters like *i* and *j*. If two elements from the same sentence are both indexed with say, *i*, it means they **corefer** (have same referent). If they are indexed with, say, *i* and *j*, we assume they have different referents.

(20) Jo_{*i*} said that they_{*i*}... Coreferring R-expression and pronoun

(21) He_{*i*} told himself_{*i*}... Coreferring pronoun and anaphor

(22) She_{*i*} told her_{*j*}... Non-coreferring pronouns

- Under which syntactic configuration(s) can pronouns, anaphors, and R-expressions corefer?

When to use which pronoun

- Let us further investigate the syntactic conditions governing the distribution of:
 - personal pronouns (now just “**pronouns**”);
 - reflexives/reciprocals (called “**anaphors**”);
 - as well as referring expressions (“**R-expressions**”)—typically proper names and full DPs.
- These elements will be indexed with letters like *i* and *j*. If two elements from the same sentence are both indexed with say, *i*, it means they **corefer** (have same referent). If they are indexed with, say, *i* and *j*, we assume they have different referents.

(20) Jo_{*i*} said that they_{*j*}... Coreferring R-expression and pronoun

(21) He_{*i*} told himself_{*j*}... Coreferring pronoun and anaphor

(22) She_{*i*} told her_{*j*}... Non-coreferring pronouns

- Under which syntactic configuration(s) can pronouns, anaphors, and R-expressions corefer?

When to use which pronoun

- Let us further investigate the syntactic conditions governing the distribution of:
 - personal pronouns (now just “**pronouns**”);
 - reflexives/reciprocals (called “**anaphors**”);
 - as well as referring expressions (“**R-expressions**”)—typically proper names and full DPs.
- These elements will be indexed with letters like *i* and *j*. If two elements from the same sentence are both indexed with say, *i*, it means they **corefer** (have same referent). If they are indexed with, say, *i* and *j*, we assume they have different referents.

(20) Jo_{*i*} said that they_{*i*}... Coreferring R-expression and pronoun

(21) He_{*i*} told himself_{*i*}... Coreferring pronoun and anaphor

(22) She_{*i*} told her_{*j*}... Non-coreferring pronouns

- Under which syntactic configuration(s) can pronouns, anaphors, and R-expressions corefer?

The distribution of R-expressions

- (23) a. **Ed_i** likes *him_i/him_j/himself_i
b. *He_i/He_j/*Himself_i likes **Ed_i**

- (23a) shows R-expressions can come first, as soon as they are not “followed” by a coreferring pronoun.
- (23b) shows R-expressions don't like to come “after” a coreferring element, whether it's a pronoun or an anaphor.

The distribution of R-expressions

- (23) a. **Ed_i** likes *him_i/him_j/himself_i
b. *He_i/He_j/*Himself_i likes **Ed_i**

- (23a) shows R-expressions can come first, as soon as they are not “followed” by a coreferring pronoun.
- (23b) shows R-expressions don't like to come “after” a coreferring element, whether it's a pronoun or an anaphor.

The distribution of R-expressions

- (23) a. **Ed_i** likes *him_i/him_j/himself_i
b. *He_i/He_j/*Himself_i likes **Ed_i**

- (23a) shows R-expressions can come first, as soon as they are not “followed” by a coreferring pronoun.
- (23b) shows R-expressions don’t like to come “after” a coreferring element, whether it’s a pronoun or an anaphor.

The distribution of pronouns

- (24) a. **He_i** likes *Ed_i/him_j/himself_i
b. *Ed_i/He_j/*Himself_i likes **him_i**

- (24a) shows pronouns can come first, as soon as they are not “followed” by a coreferring R-expression.
- (24b) shows pronouns don’t like to come “after” a coreferring element, whether it’s a pronoun or an anaphor.
- Pronouns and R-expressions seem to follow similar constraints in simple sentences.

The distribution of pronouns

- (24) a. **He_i** likes *Ed_i/him_j/himself_i
b. *Ed_i/He_j/*Himself_i likes **him_i**

- (24a) shows pronouns can come first, as soon as they are not “followed” by a coreferring R-expression.
- (24b) shows pronouns don’t like to come “after” a coreferring element, whether it’s a pronoun or an anaphor.
- Pronouns and R-expressions seem to follow similar constraints in simple sentences.

The distribution of pronouns

- (24) a. **He_i** likes *Ed_i/him_j/himself_i
b. *Ed_i/He_j/*Himself_i likes **him_i**

- (24a) shows pronouns can come first, as soon as they are not “followed” by a coreferring R-expression.
- (24b) shows pronouns don’t like to come “after” a coreferring element, whether it’s a pronoun or an anaphor.
- Pronouns and R-expressions seem to follow similar constraints in simple sentences.

The distribution of pronouns

- (24) a. **He_i** likes *Ed_i/him_j/himself_i
b. *Ed_i/He_j/*Himself_i likes **him_i**

- (24a) shows pronouns can come first, as soon as they are not “followed” by a coreferring R-expression.
- (24b) shows pronouns don’t like to come “after” a coreferring element, whether it’s a pronoun or an anaphor.
- Pronouns and R-expressions seem to follow similar constraints in simple sentences.

The distribution of anaphors

- (25) a. **Himself_i** likes *Ed_i/*him_i/*him_j
b. Ed_i/He_j/*He_j likes **himself_i**

- Anaphors (above, reflexive anaphors) don't like to come "first" – whatever "follows".
- When they come "second" they need to be "preceded" by a coreferring expression, R-expression or pronoun.
- The examples below show this also applies to reciprocals (not only reflexives).

- (26) a. **Each other_i** like *Ed and Al_i/*them_i/*them_j
b. Ed and Al_i/They_j/*They_j like **each other_i**

The distribution of anaphors

- (25) a. **Himself_i** likes *Ed_i/*him_i/*him_j
b. Ed_i/He_j/*He_j likes **himself_i**

- Anaphors (above, reflexive anaphors) don't like to come "first" – whatever "follows".
- When they come "second" they need to be "preceded" by a coreferring expression, R-expression or pronoun.
- The examples below show this also applies to reciprocals (not only reflexives).

- (26) a. **Each other_i** like *Ed and Al_i/*them_i/*them_j
b. Ed and Al_i/They_j/*They_j like **each other_i**

The distribution of anaphors

- (25) a. **Himself_i** likes *Ed_i/*him_i/*him_j
b. Ed_i/He_i/*He_j likes **himself_i**

- Anaphors (above, reflexive anaphors) don't like to come "first" – whatever "follows".
- When they come "second" they need to be "preceded" by a coreferring expression, R-expression or pronoun.
- The examples below show this also applies to reciprocals (not only reflexives).

- (26) a. **Each other_i** like *Ed and Al_i/*them_i/*them_j
b. Ed and Al_i/They_i/*They_j like **each other_i**

The distribution of anaphors

- (25) a. **Himself_i** likes *Ed_i/*him_i/*him_j
b. Ed_i/He_i/*He_j likes **himself_i**

- Anaphors (above, reflexive anaphors) don't like to come "first" – whatever "follows".
- When they come "second" they need to be "preceded" by a coreferring expression, R-expression or pronoun.
- The examples below show this also applies to reciprocals (not only reflexives).

- (26) a. **Each other_i** like *Ed and Al_i/*them_i/*them_j
b. Ed and Al_i/They_i/*They_j like **each other_i**

Summary of the intuitions, and a question

- R-expressions and pronouns don't like to be “preceded” by a coreferring expression, while anaphors need to be.
- But what kind of precedence is this really? Sentences (23-25) suggest it could be linear order.
- In (27) however, *Ed* linearly precedes a coreferring R-expression and pronoun, yet both are fine. Additionally, *Ed* precedes a coreferring anaphor, but this anaphor remains unhappy! The pattern seems completely reversed. What should we conclude?

(27) *Ed_i's mom_j likes Ed_i/him_i/*himself_i*

- It seems like **linear precedence is not the right concept**. Let's draw trees to see if hierarchical structure can help make sense of both (27) and (23-25).

Summary of the intuitions, and a question

- R-expressions and pronouns don't like to be “preceded” by a coreferring expression, while anaphors need to be.
- But what kind of precedence is this really? Sentences (23-25) suggest it could be linear order.
- In (27) however, *Ed* linearly precedes a coreferring R-expression and pronoun, yet both are fine. Additionally, *Ed* precedes a coreferring anaphor, but this anaphor remains unhappy! The pattern seems completely reversed. What should we conclude?

(27) Ed_i 's mom_j likes Ed_i /him_i/*himself_i

- It seems like **linear precedence is not the right concept**. Let's draw trees to see if hierarchical structure can help make sense of both (27) and (23-25).

Summary of the intuitions, and a question

- R-expressions and pronouns don't like to be “preceded” by a coreferring expression, while anaphors need to be.
- But what kind of precedence is this really? Sentences (23-25) suggest it could be linear order.
- In (27) however, *Ed* linearly precedes a coreferring R-expression and pronoun, yet both are fine. Additionally, *Ed* precedes a coreferring anaphor, but this anaphor remains unhappy! The pattern seems completely reversed. What should we conclude?

(27) *Ed_i's mom_j likes Ed_i/him_i/*himself_i*

- It seems like **linear precedence is not the right concept**. Let's draw trees to see if hierarchical structure can help make sense of both (27) and (23-25).

Summary of the intuitions, and a question

- R-expressions and pronouns don't like to be “preceded” by a coreferring expression, while anaphors need to be.
- But what kind of precedence is this really? Sentences (23-25) suggest it could be linear order.
- In (27) however, *Ed* linearly precedes a coreferring R-expression and pronoun, yet both are fine. Additionally, *Ed* precedes a coreferring anaphor, but this anaphor remains unhappy! The pattern seems completely reversed. What should we conclude?

(27) Ed_i 's mom_j likes Ed_i /him_i/*himself_i

- It seems like linear precedence is not the right concept. Let's draw trees to see if hierarchical structure can help make sense of both (27) and (23-25).

Summary of the intuitions, and a question

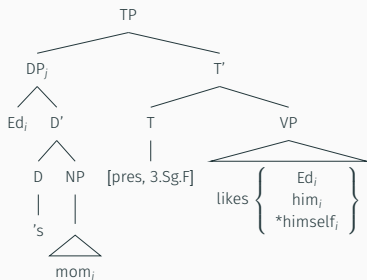
- R-expressions and pronouns don't like to be “preceded” by a coreferring expression, while anaphors need to be.
- But what kind of precedence is this really? Sentences (23-25) suggest it could be linear order.
- In (27) however, *Ed* linearly precedes a coreferring R-expression and pronoun, yet both are fine. Additionally, *Ed* precedes a coreferring anaphor, but this anaphor remains unhappy! The pattern seems completely reversed. What should we conclude?

(27) Ed_i 's mom_j likes Ed_i /him_i/*himself_i

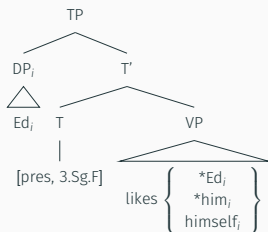
- It seems like **linear precedence is not the right concept**. Let's draw trees to see if hierarchical structure can help make sense of both (27) and (23-25).

Hierarchical precedence

(27) Ed_i 's mom_j likes Ed_i /him_i/*himself_i



(28) Ed_i likes * Ed_i /*him_i/himself_i



- R-expressions and pronouns seem fine when the coreferring antecedent is “buried” inside a non-coreferring DP.
- Anaphors are unhappy with “buried” antecedents; structurally, they need the coreferring antecedent to be the sister of an element dominating them (here T').

Defining binding

- The relation we have informally defined to spell out anaphors' licensing conditions is called **C-Command**.
- X C-Command Y if X is the sister of a node dominating Y.¹¹ To recap:
 - R-expressions and pronouns don't like to be C-Commanded by a coreferring expression (e.g. *Ed_i*), though they are fine being C-Commanded by a non-coreferring expression (e.g. *Ed_i's mother_j*).
 - Anaphors need to be C-Commanded by a coreferring expression (e.g. *Ed_i*).
- The conjunction of coreference and C-Command is called **binding**. An element is **bound** if C-Commanded by a coreferring expression; it's **free** if not bound.¹²

¹¹A node Z dominates a node Y if Z is the root of a subtree containing Y.

¹²I.e. Not C-Commanded by anything, or C-Commanded by something non-coreferring.

Defining binding

- The relation we have informally defined to spell out anaphors' licensing conditions is called **C-Command**.
- X C-Command Y if X is the sister of a node dominating Y.¹¹ To recap:
 - R-expressions and pronouns don't like to be C-Commanded by a coreferring expression (e.g. *Ed_i*), though they are fine being C-Commanded by a non-coreferring expression (e.g. *Ed_i's mother_j*).
 - Anaphors need to be C-Commanded by a coreferring expression (e.g. *Ed_i*).
- The conjunction of coreference and C-Command is called **binding**. An element is **bound** if C-Commanded by a coreferring expression; it's **free** if not bound.¹²

¹¹A node Z dominates a node Y if Z is the root of a subtree containing Y.

¹²I.e. Not C-Commanded by anything, or C-Commanded by something non-coreferring.

Defining binding

- The relation we have informally defined to spell out anaphors' licensing conditions is called **C-Command**.
- X C-Command Y if X is the sister of a node dominating Y.¹¹ To recap:
 - R-expressions and pronouns don't like to be C-Commanded by a coreferring expression (e.g. *Ed_i*), though they are fine being C-Commanded by a non-coreferring expression (e.g. *Ed_i's mother_j*).
 - Anaphors need to be C-Commanded by a coreferring expression (e.g. *Ed_i*).
- The conjunction of coreference and C-Command is called **binding**. An element is **bound** if C-Commanded by a coreferring expression; it's **free** if not bound.¹²

¹¹A node Z dominates a node Y if Z is the root of a subtree containing Y.

¹²I.e. Not C-Commanded by anything, or C-Commanded by something non-coreferring.

Defining binding

- The relation we have informally defined to spell out anaphors' licensing conditions is called **C-Command**.
- X C-Command Y if X is the sister of a node dominating Y.¹¹ To recap:
 - R-expressions and pronouns don't like to be C-Commanded by a coreferring expression (e.g. *Ed_i*), though they are fine being C-Commanded by a non-coreferring expression (e.g. *Ed_i's mother_j*).
 - Anaphors need to be C-Commanded by a coreferring expression (e.g. *Ed_i*).
- The conjunction of coreference and C-Command is called **binding**. An element is **bound** if C-Commanded by a coreferring expression; it's **free** if not bound.¹²

¹¹A node Z dominates a node Y if Z is the root of a subtree containing Y.

¹²I.e. Not C-Commanded by anything, or C-Commanded by something non-coreferring.

Defining binding

- The relation we have informally defined to spell out anaphors' licensing conditions is called **C-Command**.
- X C-Command Y if X is the sister of a node dominating Y.¹¹ To recap:
 - R-expressions and pronouns don't like to be C-Commanded by a coreferring expression (e.g. *Ed_i*), though they are fine being C-Commanded by a non-coreferring expression (e.g. *Ed_i's mother_j*).
 - Anaphors need to be C-Commanded by a coreferring expression (e.g. *Ed_i*).
- The conjunction of coreference and C-Command is called **binding**. An element is **bound** if C-Commanded by a coreferring expression; it's **free** if not bound.¹²

¹¹A node Z dominates a node Y if Z is the root of a subtree containing Y.

¹²I.e. Not C-Commanded by anything, or C-Commanded by something non-coreferring.

Binding Principles (first attempt to be revised)

- Anaphors need to be **bound**, i.e. C-Commanded by a coreferring expression.
- R-expressions and pronouns should be **free**, i.e. not C-Commanded by a coreferring expression.
- The previous point suggests R-expressions like proper names and pronouns like *he* or *she* should distribute exactly the same, which intuitively may not sound right! You might feel that pronouns do not need to be free *simpliciter*. Let's check that.

Binding Principles (first attempt to be revised)

- Anaphors need to be **bound**, i.e. C-Commanded by a coreferring expression.
- R-expressions and pronouns should be **free**, i.e. not C-Commanded by a coreferring expression.
- The previous point suggests R-expressions like proper names and pronouns like *he* or *she* should distribute exactly the same, which intuitively may not sound right! You might feel that pronouns do not need to be free *simpliciter*. Let's check that.

Binding Principles (first attempt to be revised)

- Anaphors need to be **bound**, i.e. C-Commanded by a coreferring expression.
- R-expressions and pronouns should be **free**, i.e. not C-Commanded by a coreferring expression.
- The previous point suggests R-expressions like proper names and pronouns like *he* or *she* should distribute exactly the same, which intuitively may not sound right! You might feel that pronouns do not need to be free *simpliciter*. Let's check that.

(29) Ed_i said that Jo_j likes $*Ed_i/him_i$.

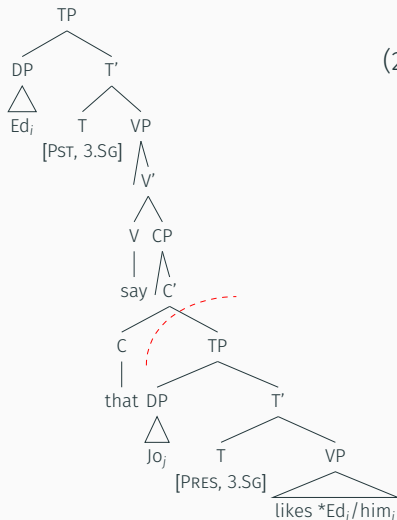
- In (29), the R-expression *Ed* is banned in the embedded clause, while the pronoun *him* is fine, even when coreferential with *Ed* in the main (=matrix) sentence.
- This suggests R-expressions and pronouns have slightly different distributions! Specifically that pronouns *can* be C-Commanded by a coreferring expression (i.e. bound) if **that expression is “far”/“high” enough**. R-expressions by contrast, need to be free *simpliciter*.

Local vs. non-local binding

(29) Ed_i said that Jo_j likes $*Ed_i/him_i$.

- In (29), the R-expression *Ed* is banned in the embedded clause, while the pronoun *him* is fine, even when coreferential with *Ed* in the main (=matrix) sentence.
- This suggests R-expressions and pronouns have slightly different distributions! Specifically that pronouns *can* be C-Commanded by a coreferring expression (i.e. bound) **if that expression is “far”/“high” enough**. R-expressions by contrast, need to be free *simpliciter*.

Local vs. non-local binding

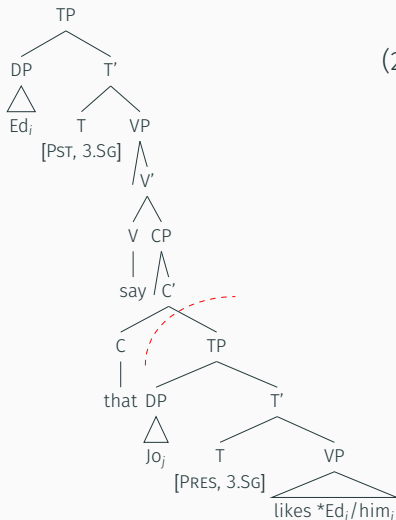


(29) Ed_i said that Jo_j likes *Ed_i/him_i.

- We define the **local domain** of a pronoun (or anaphor) as the smallest TP dominating it.
- In (29) the local domain of *him_i* is the embedded TP. This TP does not contain *him_i*'s binder Ed_i, making *him_i* locally free.¹³
- We could therefore assume pronouns like *him_i* must be free in their local domain.

¹³Jo_j, though it C-Commands *him_i* locally, does not corefer with it, i.e. no binding.

Local vs. non-local binding

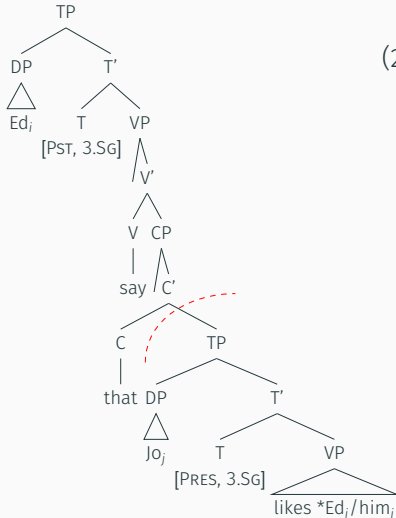


(29) Ed_i said that Jo_j likes $*Ed_i/him_i$.

- We define the **local domain** of a pronoun (or anaphor) as the smallest TP dominating it.
- In (29) the local domain of him_i is the embedded TP. This TP does not contain him_i 's binder Ed_i , making him_i locally free.¹³
- We could therefore assume pronouns like him_i must be free in their local domain.

¹³ Jo_j , though it C-Commands him_i locally, does not corefer with it, i.e. no binding.

Local vs. non-local binding



(29) Ed_i said that Jo_j likes *Ed_i/him_i.

- We define the **local domain** of a pronoun (or anaphor) as the smallest TP dominating it.
- In (29) the local domain of *him_i* is the embedded TP. This TP does not contain *him_i*'s binder Ed_i, making *him_i* locally free.¹³
- We could therefore assume pronouns like *him_i* must be free in their local domain.

¹³Jo_j, though it C-Commands *him_i* locally, does not corefer with it, i.e. no binding.

What about anaphors?

- Let's use the same nested structure, but replace *him_i* with anaphors.

(30) Ed_i said that Jo_j likes **himself_i*/*herself_j*

- We observe that both *himself_i* and *herself_j* are bound here, resp. by Ed_i in the matrix and Jo_j in the embedded clause. Yet, only *herself_j* is happy!
- This suggests **anaphors are also locally constrained**; specifically, they must be locally bound.
- In (30), Jo_j binds *herself_j* in this anaphor's local domain, while Ed_i is too high to bind *himself_i* in its local domain.

What about anaphors?

- Let's use the same nested structure, but replace *him_i* with anaphors.

(30) Ed_i said that Jo_j likes * $himself_i$ / $herself_j$

- We observe that both *himself_i* and *herself_j* are bound here, resp. by Ed_i in the matrix and Jo_j in the embedded clause. Yet, only *herself_j* is happy!
- This suggests **anaphors are also locally constrained**; specifically, they must be locally bound.
- In (30), Jo_j binds *herself_j* in this anaphor's local domain, while Ed_i is too high to bind *himself_i* in its local domain.

What about anaphors?

- Let's use the same nested structure, but replace *him_i* with anaphors.

(30) Ed_i said that Jo_j likes **himself_i*/*herself_j*

- We observe that both *himself_i* and *herself_j* are bound here, resp. by Ed_i in the matrix and Jo_j in the embedded clause. Yet, only *herself_j* is happy!
- This suggests **anaphors are also locally constrained**; specifically, they must be locally bound.
- In (30), Jo_j binds *herself_j* in this anaphor's local domain, while Ed_i is too high to bind *himself_i* in its local domain.

What about anaphors?

- Let's use the same nested structure, but replace *him_i* with anaphors.

(30) Ed_i said that Jo_j likes **himself_i* / *herself_j*

- We observe that both *himself_i* and *herself_j* are bound here, resp. by Ed_i in the matrix and Jo_j in the embedded clause. Yet, only *herself_j* is happy!
- This suggests **anaphors are also locally constrained**; specifically, they must be locally bound.
- In (30), Jo_j binds *herself_j* in this anaphor's local domain, while Ed_i is too high to bind *himself_i* in its local domain.

- The local domain of an expression is the smallest TP dominating it.
- **Principle A:** Anaphors need to be **bound**, i.e. C-Commanded by a coreferring expression, within their local domain.
- **Principle B:** Pronouns need to be **free** within their local domain.
- **Principle C:** R-expressions should be **free**, i.e. not C-Commanded by a coreferring expression at any level.

- The local domain of an expression is the smallest TP dominating it.
- **Principle A:** Anaphors need to be **bound**, i.e. C-Commanded by a coreferring expression, within their local domain.
- **Principle B:** Pronouns need to be **free** within their local domain.
- **Principle C:** R-expressions should be **free**, i.e. not C-Commanded by a coreferring expression at any level.

- The local domain of an expression is the smallest TP dominating it.
- **Principle A:** Anaphors need to be **bound**, i.e. C-Commanded by a coreferring expression, within their local domain.
- **Principle B:** Pronouns need to be **free** within their local domain.
- **Principle C:** R-expressions should be **free**, i.e. not C-Commanded by a coreferring expression at any level.

- The local domain of an expression is the smallest TP dominating it.
- **Principle A:** Anaphors need to be **bound**, i.e. C-Commanded by a coreferring expression, within their local domain.
- **Principle B:** Pronouns need to be **free** within their local domain.
- **Principle C:** R-expressions should be **free**, i.e. not C-Commanded by a coreferring expression at any level.

Exercises

- Explain, based on the 3 Binding Principles:

- (31)
- a. Ed said Jo likes her
 - b. Ed said he likes himself
 - c. * Ed said himself likes him
 - d. Ed wants to like *him/himself
 - e. Ed wants Jo to like him/*himself
 - f. Which aspect of *him/himself does Ed dislike?

References i



Geach, P. T. (1962). *Reference and generality*. Cornell University Press.



Evans, G. (1977). Pronouns, quantifiers, and relative clauses. *Canadian Journal of Philosophy*, 7(3), 467–536.



Chomsky, N. (1981). *Lectures on government and binding*. Foris Publications.



Heim, I. (1990). E-type pronouns and donkey anaphora. *Linguistics and Philosophy*, 13(2), 137–177. <https://doi.org/10.1007/BF00627309>



Reinhart, T., & Reuland, E. (1993). Reflexivity. *Linguistic Inquiry*, 24(4), 657–720.



Cardinaletti, A., & Starke, M. (1994). The typology of structural deficiency: On the three grammatical classes. *Working Papers in Linguistics*, 4, 41–109.



Landau, I. (2000). *Elements of control: Structure and meaning in infinitival constructions*. Kluwer Academic Publishers.



Schlenker, P. (2003). A plea for monsters. *Linguistics and Philosophy*, 26(3), 279–310. <https://doi.org/10.1023/A:1024694329542>



Landau, I. (2004). The scale of finiteness and the calculus of control. *Natural Language & Linguistic Theory*, 22(4), 811–877. <https://doi.org/10.1007/s11049-004-4265-5>



von Fintel, K. (2004). **Would you believe it? the king of france is back! (presuppositions and truth-value intuitions)**. In E. Georgala et al. (Eds.), *Proceedings of the 13th semantics and linguistic theory conference* (pp. 269–286).



Harbour, D., Adger, D., & Béjar, S. (Eds.). (2008). ***Phi theory: Phi-features across modules and interfaces***. Oxford University Press.
<https://doi.org/10.1093/oso/9780199213764.001.0001>



Heim, I. (2008). **Features on bound pronouns**. In D. Harbour, D. Adger, & S. Béjar (Eds.), *Phi theory: Phi-features across modules and interfaces* (pp. 35–56). Oxford University Press. <https://doi.org/10.1093/oso/9780199213764.003.0002>



Sauerland, U. (2008). **On the semantic markedness of phi-features**. In D. Harbour, D. Adger, & S. Béjar (Eds.), *Phi theory: Phi-features across modules and interfaces* (pp. 37–60). Oxford University Press.



Shklovsky, K., & Sudo, Y. (2014). **The syntax of monsters**. *Linguistic Inquiry*, 45(3), 381–402. https://doi.org/10.1162/LING_a_00160



Wang, R. (2023). **Honorifics without [hon]**. *Natural Language & Linguistic Theory*, 41(3), 1287–1347. <https://doi.org/10.1007/s11049-022-09563-0>